Name
Date
Course

Practice Intermediate Algebra Test III (Version 2)

I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
II) Show all work.
III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for this test.
V) You do not have to write the Roman numeral for each exponent rule that you use on this test.
VI) Do not leave your answers with fraction exponents, negative exponents, zero power exponents, or 1st power exponents.
VII) Each problem is worth four points for a total of 100 points.

Simplify.

① $\sqrt{4^5}$

② $\left(b^{\frac{1}{5}}\right)^5$

③ $9^{-2} \cdot 9^{\frac{5}{2}}$

④ $\left(\frac{4}{9}\right)^{-2}$

⑤ $\dfrac{a^{-5}b^{-1}}{(-2)^{-2}x^{-4}}$

⑥ $\left(\sqrt[3]{\dfrac{125^6}{125^2}}\right)^{-\frac{1}{2}}$

⑦ $\left(\dfrac{b^{12} \cdot x^0}{x^3 b^9}\right)^{-1}$

Find the values of the following logarithms without a calculator.

⑧ $\log_2 \sqrt[3]{2}$

⑨ $\log_{37} 37$

⑩ If $\log_k 4 \approx 0.7$, use this information to approximate the following value.

$\log_k\left(\frac{1}{16}\right)$

Solve.

⑪ $3^x \cdot 2^x = \frac{1}{36}$

⑫ $4^{\log_4 x + 2} = 10$

Solve. If an answer is irrational, round to the nearest ten thousandth.

⑬ $\frac{2}{3} = \log_8(-5x+6) + \frac{1}{3}$

⑭ $4(3^{1-2x}) + 2 = 9$

(15) $3(2^{2x}) - 6(2^{x}) + 1 = 0$

(16) $\frac{1}{125} = \sqrt[3]{625^{x}}$

(17) $\log_3(x+4) + \log_3(x-2) = 3$

Write your answer as an exact expression.

18. $1 = 5\log_x 2 - \log_x 3$

19. $2\log_{100} X = \log_{100} 86 - \log_{100} 2$

20. $\log_4 X + \log_4(12x) - \log_4 3 = 2$

(21) $\log_4(x-1) - \log_4(x+1) = 1$

(22) $343^{2x-1} = \sqrt[3]{49}$

(23) $0 = \log_2[\log_5(3x+1) - 2]$

Simplify.

24 $\left(x^{\frac{1}{2}}y \cdot x^{\frac{1}{2}}\right)^3\left(b^2 \cdot x^0 \cdot y^5 \cdot y^2\right)^{\frac{1}{2}}$

25 $\left(\frac{3b^2y^4}{2y} \div \frac{1}{b^{-5}y^{\frac{3}{4}}} \cdot \frac{2}{3}\right)^{-1} \cdot \left(\frac{x^2y^4}{y^3x^2}\right)^2 \cdot \frac{(x^3y)^2}{(x^2y^4)^{-3}}$

Answers

1. $\boxed{32}$
2. $\boxed{b}$
3. $\boxed{3}$
4. $\boxed{\frac{81}{16}}$
5. $\boxed{\frac{4x^4}{a^5 b}}$
6. $\boxed{\frac{1}{25}}$
7. $\boxed{\frac{x^3}{b^3}}$
8. $\boxed{\frac{1}{3}}$
9. $\boxed{1}$
10. $\boxed{-1.4}$
11. $\boxed{x=-2}$
12. $\boxed{x=\frac{5}{8}}$
13. $\boxed{x=\frac{4}{5}}$
14. $\boxed{x \approx 0.2453}$
15. $\boxed{x \approx -2.4461 \text{ or } x \approx 0.8612}$
16. $\boxed{x=-\frac{9}{4}}$
17. $\boxed{x=5,\ x \neq -7}$
18. $\boxed{x=\frac{32}{3}}$
19. $\boxed{x=\sqrt{43},\ x \neq -\sqrt{43}}$
20. $\boxed{x=2,\ x \neq -2}$
21. $\boxed{\text{No solution, } x \neq -1}$
22. $\boxed{x=\frac{11}{18}}$
23. $\boxed{x=41\frac{1}{3} \text{ or } 41.\overline{3} \text{ or } \frac{124}{3}}$
24. $\boxed{bx^3\sqrt{y^{13}}}$
25. $\boxed{x^{12}b^3\sqrt[4]{y^{49}}}$